

SOC L1 INCIDENT INVESTIGATION

Case 02 - Brute-Force / Password Attack Followed by Successful Authentication

Field	Case Detail
Portfolio Owner	Akachi Maduako
Environment	SOC Lab / Synthetic Dataset
Analyst Level	L1 triage with L2-style correlation and scoping
Final Disposition	TRUE POSITIVE - HIGH SEVERITY - ESCALATE
Primary Account	j.smith
Suspicious Source	185.220.101.45
Affected Hosts	WIN-LT-023; WIN-SRV-02

1. Executive Summary

A synthetic Windows authentication dataset was investigated after repeated failed logons were observed from **185.220.101.45**. The source attempted authentication against multiple accounts and generated **18 failed logons against j.smith**. The same source subsequently produced a successful remote-interactive logon to **WIN-LT-023**, followed seconds later by a special-privileges event and an explicit-credentials event. Approximately 14 minutes later, the same account and source IP successfully authenticated to **WIN-SRV-02** using a network logon with NTLM.

The evidence supports a high-confidence account-compromise scenario following a password attack. Secondary-host access is suspicious and warrants investigation; however, the available dataset does not contain sufficient post-authentication telemetry to claim confirmed lateral movement.

2. Investigation Objective

Determine whether the failed authentication activity represents benign user error or a malicious password attack; identify whether an account was successfully accessed; establish affected accounts, hosts, and source; correlate related Windows security events; and determine whether escalation is required.

3. Evidence Reviewed

Primary evidence: **evidence/windows_security_events.csv** from the synthetic SOC lab dataset. Analysis focused on Windows Security Event IDs **4625, 4624, 4672, and 4648** and correlated activity by source IP, account, timestamp, host, logon type, process, authentication package, and workstation name.

4. Investigation Methodology

Step 1 - Identify the dominant failed-login source. Failed Event ID 4625 records were grouped by SourceIP. 185.220.101.45 produced 27 failures, the largest observed count.

Step 2 - Determine targeted accounts. Filtering failed logons from 185.220.101.45 showed j.smith with 18 failures, followed by a.chen, m.patel, and s.williams with 3 failures each. This multi-account pattern is consistent with password spraying/brute-force behavior.

Step 3 - Search for successful authentication. Event ID 4624 records revealed successful logons for j.smith from the same source IP.

Step 4 - Pivot on account and host. Events involving j.smith and/or 185.220.101.45 were placed into a chronological timeline, exposing closely timed 4624, 4672, and 4648 events on WIN-LT-023 and a later 4624 on WIN-SRV-02.

Step 5 - Scope the second host. WIN-SRV-02 was searched from 10:30Z to 10:45Z. The dataset contained one matching event: a successful Type 3 network logon for j.smith from 185.220.101.45 at 10:36:11Z. No additional post-authentication events were present in that time window.

5. Key Evidence

Time (UTC)	Event	Account	Source / Destination	Significance
Before 10:22	4625	Multiple	185.220.101.45	Repeated failures across accounts; 18 against j.smith
10:22:11	4624	j.smith	185.220.101.45 -> WIN-LT-023	Successful Logon; Type 10; winlogon.exe; Negotiate
10:22:20	4672	j.smith	WIN-LT-023	Special privileges assigned to new logon; lsass.exe
10:22:44	4648	j.smith	WIN-LT-023	Explicit credentials used; svchost.exe
10:36:11	4624	j.smith	185.220.101.45 -> WIN-SRV-02	Successful Type 3 network logon; services.exe; NTLM

6. Attack Timeline

Phase 1 - Password attack: 185.220.101.45 generated repeated Event ID 4625 failures against several accounts. j.smith was the most heavily targeted with 18 failures.

Phase 2 - Successful remote authentication: At 10:22:11Z, Event ID 4624 recorded j.smith successfully logging onto WIN-LT-023 from 185.220.101.45. The record shows Logon Type 10, winlogon.exe, and Negotiate.

Phase 3 - Privileged/credential-related activity: At 10:22:20Z, Event ID 4672 recorded special privileges assigned to the new logon. At 10:22:44Z, Event ID 4648 recorded an explicit-credentials authentication attempt associated with j.smith on WIN-LT-023.

Phase 4 - Secondary-host authentication: At 10:36:11Z, Event ID 4624 recorded j.smith authenticating from 185.220.101.45 to WIN-SRV-02 using Logon Type 3 and NTLM. This is suspicious in context but is classified as suspected secondary-host/lateral access rather than confirmed lateral movement.

7. Analyst Findings

Finding 1: The activity is not consistent with a single mistyped password. One source generated failures against multiple accounts, concentrated against j.smith.

Finding 2: The suspicious source subsequently authenticated successfully as j.smith, strongly increasing the likelihood of credential compromise.

Finding 3: Special-privilege and explicit-credential events occurred within seconds of the WIN-LT-023 logon. The dataset does not expose the exact privilege list or the target of the explicit-credential event, so those details are not inferred.

Finding 4: j.smith later authenticated to WIN-SRV-02 from the same suspicious source. This warrants escalation as possible secondary-host access. The evidence does not prove lateral movement or malicious post-authentication execution on WIN-SRV-02.

8. Disposition and Escalation

Disposition: TRUE POSITIVE

Severity: HIGH

Action: ESCALATE TO L2 / INCIDENT RESPONSE

Escalation is justified because repeated multi-account failures were followed by successful authentication of the most

heavily targeted account, closely timed privilege/credential-related events, and later authentication to a second system.

9. Recommended Response Actions

1. Validate j.smith activity with the account owner and disable or restrict the account if unauthorized.
2. Reset affected credentials and revoke active sessions/tokens as appropriate.
3. Block or investigate 185.220.101.45 at relevant security controls, subject to policy.
4. Investigate and, if warranted, isolate WIN-LT-023 while endpoint telemetry is reviewed.
5. Investigate WIN-SRV-02 for process creation, services, scheduled tasks, remote administration, file activity, and other post-authentication behavior.
6. Review authentication telemetry for other accounts targeted by the source.
7. Preserve relevant logs and escalate the correlated timeline and IOCs to L2/IR.

10. Indicators and Scope

Indicator Type	Value	Assessment
Suspicious source IP	185.220.101.45	Associated with failed and successful authentication
Primary account	j.smith	Likely compromised / requires immediate validation
Primary endpoint	WIN-LT-023	Successful Type 10 authentication plus correlated events
Secondary system	WIN-SRV-02	Successful Type 3 / NTLM authentication; investigate further
Workstation identifier	ATTACK-01	Recorded in correlated authentication events
Relevant Event IDs	4625, 4624, 4672, 4648	Failure, success, special privileges, explicit credentials

11. MITRE ATT&CK; Mapping

Technique	Assessment	Evidence Basis
T1110 - Brute Force	Supported	Repeated failed authentication from one source against multiple accounts
T1078 - Valid Accounts	Supported / suspected compromise	Successful authentication as j.smith after repeated failures
T1021 - Remote Services	Contextual / suspected	Type 10 remote-interactive logon; dataset does not explicitly identify the remote-service protocol
Lateral Movement	Not confirmed	Second-host authentication is suspicious, but post-authentication telemetry is insufficient

12. CLI / Python Analysis Used

The case was investigated with lightweight Python/CLI analysis, demonstrating that core SOC triage can be reproduced without a commercial SIEM. Tasks included grouping failed Event ID 4625 records by SourceIP, counting targeted accounts, searching Event ID 4624 successes, constructing a chronological account/IP timeline, printing complete 4624/4672/4648 records, and scoping WIN-SRV-02 within a defined time window.

```
if row["EventID"] == "4625": failed[row["SourceIP"]] += 1
if row["SourceIP"] == suspicious_ip and row["EventID"] == "4624": ...
if row["Computer"] == "WIN-SRV-02" and START <= timestamp <= END: ...
```

13. SOC Skills Demonstrated

L1: alert triage; Windows authentication-log analysis; Event ID interpretation; IOC identification; failed-vs-successful authentication correlation; account/host scoping; timeline construction; true-positive disposition; severity assessment; escalation documentation; evidence-based reporting.

L2-adjacent: pivoting across IP/account/host entities; correlating privilege and explicit-credential events; investigating secondary-host authentication; distinguishing confirmed evidence from hypotheses; identifying telemetry gaps requiring endpoint/IR follow-up.

14. Evidence Limitations

The Event 4672 record does not include the actual assigned privilege list, and Event 4648 does not provide enough target detail to identify exactly where the explicit credentials were used. The WIN-SRV-02 window contains only the successful authentication event and no subsequent process or endpoint activity. Therefore, this report does not claim confirmed privilege escalation or confirmed lateral movement.

15. Portfolio Disclosure

This is a SOC Lab / Synthetic Dataset investigation. It demonstrates investigation methodology, log analysis, analyst reasoning, escalation, and reporting. It is not presented as a production incident handled for an employer.